Real-Time Cross-Domain Gesture and User Identification via COTS WiFi

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***Abstract*—WiFi-based gesture recognition has emerged as a promising alternative to computer vision, enabling seamless integration and enhanced interaction in human-computer interaction systems. Simultaneously identifying users during gesture recognition is vital for improving security and personalization. However, existing WiFi-based dual-task recognition approaches often rely on handcrafted features, which hinder precision and introduce delays in cross-domain scenarios. To address these challenges, we propose WiDual, a real-time system for cross-domain gesture recognition and user identification using WiFi signals. By integrating spatial and channel attention mechanisms, WiDual adaptively extracts crucial features for dual-task recognition. The system employs Channel State Information (CSI) visualization to convert WiFi signals into images, facilitating efficient feature extraction and minimizing information loss and latency. Furthermore, a collaborative module fuses gesture and user identity features, enhancing recognition performance. Experimental evaluations on a public dataset with six gestures and six users across diverse environments demonstrate WiDual's effectiveness. It achieves 96% accuracy in cross-domain gesture recognition and 91.27% in user identification. Compared to state-of-the-art methods, WiDual improves user identification accuracy by 26%, gesture recognition by 8%, and reduces processing time sixfold, showcasing its potential for real-time applications.**

***Index Terms*—WiFi sensing, gesture recognition, user identification, channel state information.**

I. Introduction

GESTURE recognition plays a pivotal role in enhancing human-computer interaction by enabling intuitive and natural control mechanisms. Conventional vision-based gesture recognition approaches are hindered by issues such as poor lighting, occlusions, and privacy concerns. Recent advancements [1], [2], [3], [4], [5] in wireless sensing applications leveraging ubiquitous WiFi signals present a promising alternative, offering non-invasive, cost-effective, and adaptable solutions for precise gesture recognition across diverse applications such as health monitoring, assisted living, and virtual reality [6], [7], [8]. The fundamental principle of WiFi-based gesture recognition involves detecting the unique variations in received signals resulting from different gestures. Its contact-free and device-free nature makes it particularly valuable for scenarios requiring contactless interactions, especially highlighted during the COVID-19 pandemic.

In gesture recognition-based applications, user identification is gaining increasing importance, particularly in security-sensitive scenarios. For instance, in smart home environments, gesture-based control of various appliances provides significant convenience. However, without user identification mechanisms, there is a substantial risk of unauthorized commands being executed by intruders or children, potentially leading to property damage or personal injury. Moreover, integrating user identification with gesture recognition can enhance personalization settings, thereby improving the quality of the user experience. For example, in interactions with a smart TV, the system can intelligently recommend content based on the user's identity and record user behavior for more precise customization. As a result, recent researchers have increasingly focused on achieving dual tasks of gesture and user recognition.

Nevertheless, achieving simultaneous gesture and user recognition with high accuracy in real-time is a nontrivial task. Fig. 1 illustrates the typical workflow of WiFi-based dual-task applications, which comprises three primary phases: WiFi data collection, data preprocessing (including data denoise and feature extraction), and recognition model training and testing. During the data collection phase, receivers capture signal data that includes physical information about the signal propagation environment. This data is then preprocessed to remove unwanted noise through denoising and to identify relevant features through feature extraction. In the recognition model training and testing phase, the preprocessed data is used to train models capable of performing dual tasks of gesture and user recognition. This process faces three fundamental challenges. First, achieving low latency is difficult due to the high computational complexity involved in feature extraction. Second, WiFi signals usually contain intricate information, including environmental data,

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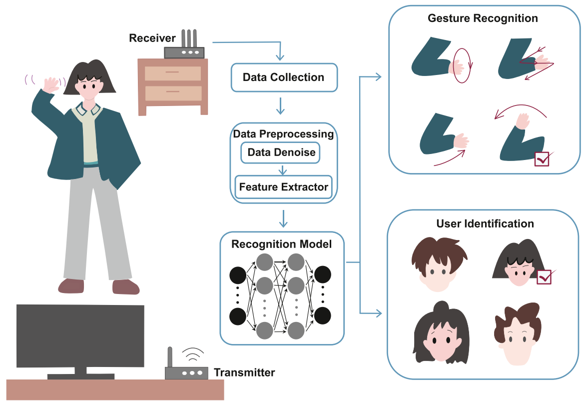


Fig. 1. The typical workflow of WiFi-based dual-task systems.

making it challenging to extract features that effectively represent both unique gesture characteristics and personalized user styles. Finally, maintaining accuracy across various domains (such as different locations, orientations, environments) is particularly challenging [4]. The same gesture performed by the same person can produce significantly different signals in different conditions due to multipath effects.

Existing research typically treats gesture recognition and user identification as separate tasks [21], [22], [23], [24], [32], [33], [34]. Recent advancements have seen the emergence of approaches utilizing WiFi for dual-task recognition. However, these dual-task approaches often fail to effectively address all three fundamental challenges [11]. WiID [12] is the pioneering system that supports both gesture recognition and user identification using WiFi signals. However, WiID falls short in facilitating cross-domain recognition, particularly across different locations and varying environments. WiGesID [9] is capable of performing gesture recognition and user identification across different domains. Nonetheless, it faces challenges in real-time performance due to its reliance on the domain-independent feature known as the body-coordinate velocity profile (BVP) [4]. This reliance introduces high computational complexity during the feature extraction process, impeding its efficiency. WiHF [10] attempts to address these limitations by introducing another domain-independent feature, the motion change pattern, to simultaneously recognize gestures and user identities. Despite this innovation, the handcrafted nature of the motion change pattern fails to capture sufficient cross-domain information. This inadequacy results in difficulties maintaining high accuracy across different domains.

In this paper, we propose WiDual, a real-time dual-task system designed for precise cross-domain gesture recognition and user identification utilizing WiFi signals. Drawing inspiration from the efficacy of deep learning in computer vision for pattern recognition, WiDual exploits a Channel State Information (CSI) visualization technique that converts WiFi signals into images for comprehensive feature extraction and model training. This approach allows WiDual to denoise and merge all CSI subcarriers into normalized time-series images. To enhance feature learning, WiDual incorporates an attention mechanism within the neural network, which autonomously identifies and extracts crucial cross-domain features relevant to both tasks. This approach mitigates the potential loss of useful information and excessive delays associated with extracting handcrafted features directly from WiFi signals. Furthermore, WiDual leverages a collaborative module to integrate gesture and user identity features, thereby improving dual-task recognition performance. We implement WiDual and evaluate it on a public dataset [4] that includes six gestures performed by six users across five different locations and five orientations in three varied environments. The results demonstrate that WiDual gains 98.58% and 98.73% accuracy for in-domain gesture recognition and user identification, respectively. In cross-domain scenarios, WiDual achieves 96% accuracy for gesture recognition and 91.27% for user identification, surpassing state-of-the-art approaches. Specifically, it delivers a 26% improvement in user identification accuracy, an 8% enhancement in gesture recognition, and a sixfold reduction in processing time compared to the state-of-the-art approach. The key contributions of this paper are as follows:

- We propose a novel dual-task network capable of performing real-time gesture and user recognition simultaneously using WiFi signals.
- We design a feature extractor based on ResNet18, which automatically mines cross-domain features of gestures and users from WiFi signals. Additionally, we introduce a collaboration module that enhances the recognition accuracy.
- We implement WiDual and conduct extensive experiments to evaluate its performance. The results validate the effectiveness and robustness of the proposed approach.

The rest of this paper is organized as follows. Section II demonstrates the problem statement. The WiDual design is detailed in Section III. Section IV introduces the implementation of the system. Evaluation and system insights are explained in Section V. We provide the discussion and future work in Section VI. Related works are reviewed in Section VII. Finally, we conclude this article in Section VIII.

II. Problem Statement

Objective. The objective of this paper is to design a WiFi-based dual-task system that can accurately and efficiently recognize gestures and user identities in a real-time manner across different domains. Specifically, we define "real-time" in WiDual as a system response time of less than 1 s. This definition aligns with widely accepted response time criteria in human-computer interaction, where a response time under 1 s is considered the upper limit for maintaining a seamless user experience without interrupting the user's cognitive flow [26]. For "cross-domain" functionality, we refer to the model's ability to generalize to testing domains that differ from the training domains. Recognizing the diversity and complexity of this problem, our study focuses on three specific cross-domain contexts: cross-orientation, cross-location, and cross-environment. In all three scenarios, the testing domains are entirely unseen during model training. For instance, in the cross-orientation scenario, the model is trained on gesture data from four directions and evaluated on data from a fifth, unseen direction. Similar procedures are applied for cross-location and cross-environment testing.

Note that, unlike RISE [36], which focuses on enhancing system robustness during deployment by addressing dynamic environmental changes (e.g., furniture movement or object removal), WiDual emphasizes pre-deployment robustness by promoting cross-domain generalization at the training stage. While WiDual and RISE tackle different challenges, their strengths are complementary, offering the potential for a more robust and adaptable sensing system.

Assumption: We assume that: ①The training dataset is double-labeled, containing labels for both gestures and user identities. ②During actual deployment, only one user performs gestures within the system's sensing range.

Dual-Task Problem Fomulation: The core idea behind using WiFi for dual tasks is that each person's gestures cause unique changes in WiFi signal propagation, detectable through WiFi's CSI. By leveraging this principle, a WiFi-based dual-task system can identify patterns from CSI signals corresponding to specific users and their behaviors. Thus, the dual-task problem can be framed as a multi-task supervised learning problem.

The dual-task learning problem involves two supervised learning tasks T_i with $i = 1, 2$. Tasks are associated with a training dataset and a testing dataset. Formally, the training dataset is represented as $D^{tr} = \{(x^i, y_1^i, y_2^i)\}_{i=1}^{n_{tr}}$, and the testing dataset is denoted as $D^{te} = \{(x^j, y_1^j, y_2^j)\}_{j=1}^{n_{te}}$. Here, x is an input instance in a d-dimensional space, y_1 is the gesture label and y_2 is the user identity label. The numbers of training and testing samples are n_{tr} and n_{te}, respectively. The primary objective of the dual-task system is to simultaneously learn both tasks, optimizing the model's performance for gesture recognition and user identification. This is achieved by finding the optimal parameters θ for the deep learning model to minimize the combined loss L across both tasks Mathematically, the optimization process is expressed as:

$$\theta^* = \min_{\theta \in \Theta = \cup_{i=1}^{n} \Theta_i} \sum_{i=1}^{N} L_i \left(\theta_i, D^{tr}\right) \tag{1}$$

L_i and Θ_i represent loss and parameters for i^{th} task. Particularly, the input data of dual-task model is x in D^{tr}, and the output is two classification results, which are the predicted gesture type and user identity. Initially, there exists a disparity between the predicted task results and the true labels, represented by $L_i(\theta_i, D^{tr})$. After assessing the loss on the training set's predictions, the subsequent step entails training the network via backpropagation and gradient descent algorithms. After multiple training iterations, the performance of the final network is ascertained by computing the loss on the test dataset.

The dual-task problem is a typical supervised machine-learning classification problem, making us use the cross-entropy loss function as the function to measure the model loss. We choose the cross-entropy loss function because it provides informative gradient signals, facilitates faster convergence by driving efficient updates to model weights, and aligns naturally with the probabilistic framework required for multi-class classification tasks. It estimates the difference between the network's actual output and the expected true class distribution. A smaller cross-entropy loss value indicates that the network's predicted class distribution is closer to the true distribution. The form of the cross-entropy loss function is as follows:

$$L(\theta, D) = \sum_{t=1}^{2} \sum_{i}^{n} log \left(1 + exp\left(-\hat{y_i}^t {y_i}^t\right)\right) \tag{2}$$

where $\hat{y_i}^t = {\theta_i}^t {x_i}^t$ represents the prediction for the ith data point x_i of the tth task, ${y_i}^t$ is the corresponding ground-truth label, θ denotes the network parameters, and D refers to the dataset including input X and label Y.

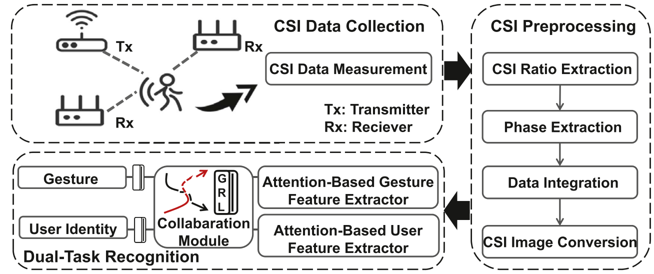


Fig. 2. The system architecture of WiDual.

III. System Design

A. System Overview

WiDual aims to achieve real-time, cross-domain recognition of user-identified gestures. As illustrated in Fig. 2, the system architecture of WiDual comprises three primary modules: CSI data collection, CSI data preprocessing, and dual-task recognition. Note that, WiDual inherently reduces the risk of exposing personal information. It focuses on extracting features relevant to gesture recognition and user identity classification directly from wireless signals, without requiring sensitive personal data such as images, audio recordings, or other identifiable biometrics.

CSI Data Collection: The first step involves gathering CSI data from WiFi signals to capture the necessary information for recognizing gestures and identifying user identities. The rationale behind this lies in the premise that diverse user behaviors manifest distinct effects on the received CSI streams, offering an opportunity to extract features or construct profiles specific to predefined behaviors. As illustrated in Fig. 2, WiDual implements a setup comprising a single Commercial Off-The-Shelf (COTS) WiFi device and multiple receivers to capture CSI data from proximate users.

CSI Data Preprocessing: This module denoises the raw CSI data and converts it into images to facilitate subsequent feature extraction and model training. Instead of extracting amplitude information, which is highly sensitive to environmental factors, we focus on extracting the more stable phase-shift information to reflect the influence of user behaviors [14]. To eliminate the time-varying random phase offset caused by the lack of synchronization between the transmitter and receiver in commodity WiFi devices, we further calculate the CSI ratio for each receiver [15], [28]. The phase change information from all CSI subcarriers is then aggregated and converted into image representations for further processing.

Dual-task Recognition: This module simultaneously performs gesture recognition and user identification using CSI images as inputs. It employs attention-based neural networks to autonomously learn and extract discriminative features critical for both tasks. To further enhance the dual-task performance, a collaboration module is implemented to merge gesture and user identity features, thereby improving the accuracy of each task through mutual feature fusion. Additionally, a Gradient Reversal Layer (GRL) is utilized to prevent interference between the tasks during the backpropagation process, ensuring that the loss from one task does not negatively impact the feature extraction and parameter optimization of the other task. This setup allows for the concurrent and efficient recognition of both gestures and user identity.

B. CSI Data Collection

WiFi Channel State Information (CSI) characterizes the attenuation and phase shifts that occur as wireless signals travel through various propagation paths. Wireless signals are transmitted from the transmitter and received by the receiver in an indoor environment. These signals take multiple paths to reach the receiver, including static paths like Line-of-Sight (LoS) and environmental reflections for walls and furniture, as well as dynamic paths influenced by human gestures. The received signal is a superposition of all these paths. The receiver's physical layer provides CSI infomation, which details the multipath propagation characteristics of the signal. Mathematically, the CSI in the frequency domain at the carrier frequency f and time t is expressed as:

$$\begin{aligned} H(f,t) &= H_s(f,t) + H_d(f,t) \\ &= H_s(f,t) + \sum_{i=1}^{n} A_i(f,t) e^{-j2\pi \frac{d_i(t)}{\lambda}} \end{aligned} \tag{3}$$

where $H_s(f,t)$ is the static component, n is the number of dynamic paths, $A_i(f,t)$, $e^{-j2\pi \frac{d_i(t)}{\lambda}}$ and $d_i(t)$ are the complex attenuation, phase shift and path length of dynamic component $H_d(f,t)$, respectively. When a person performs a gesture, the dynamic paths change, leading to variations in $d_i(t)$ and consequently the phase shift $e^{-j2\pi \frac{d_i(t)}{\lambda}}$, which alters the CSI. The same person performing the same gesture will produce similar CSI changes under the same domain, but variations in body characteristics (e.g., body thickness) among different individuals will result in distinct CSI patterns for the same gesture. By analyzing these variations in CSI phase shifts, we can infer both the specific gestures performed and the identity of the user.

C. CSI Data Preprocessing

As demonstrated in the previous section, the gesture and user identity can be portrayed by the change of phase changes in CSI. However, extracting these dynamic phase changes is complex due to errors introduced by commercial WiFi hardware, which causes random phase offsets in the raw CSI signals. These offsets include Channel Frequency Offset (CFO) and Sample Frequency Offset (SFO). The CSI affected by phase offset can be mathematically represented as:

$$H(f,t) = e^{-j\theta_{offset}} \left(H_s(f,t) + \sum_{i=1}^{n} A_i(f,t) e^{-j2\pi \frac{d_i(t)}{\lambda}} \right) \tag{4}$$

where $e^{-j\theta_{offset}}$ is a time-varying random phase offset.

For accurate gesture and identity recognition, it is crucial to eliminate this phase offset from the collected CSI data. To mitigate the phase offset problem, we use the CSI ratio method as detailed in [15], [28]. This method involves computing the ratio of CSI readings between two antennas:

$$H_r(f,t) = \frac{H_1(f,t)}{H_2(f,t)} \tag{5}$$

where $H_1(f,t)$ is the complex-valued CSI of the first antenna and $H_2(f,t)$ is that of the second antenna. The key insight is that the random phase offset remains the same across different antennas on the same WiFi Network Interface Card (NIC) because they share the same RF oscillator [27]. This allows the phase offset can be canceled out when taking the ratio of CSI readings from two antennas. Note that when a person moves their hand over a short distance, the difference in reflection path lengths between two nearby antennas $d_2(t) - d_1(t)$ can be treated as a constant. This assumption holds because the antennas are close enough that the variations in path lengths due to the movement are minimal. Therefore, the CSI ratio can be simplified and rewritten as:

$$\begin{aligned} H_r(f,t) &= \frac{H_{s,1} + A_1 e^{-j2\pi \frac{d_1(t)}{\lambda}}}{H_{s,2} + A_2 e^{-j2\pi \frac{d_2(t)-d_1(t)}{\lambda}} e^{-j2\pi \frac{d_1(t)}{\lambda}}} \\ &= \frac{\mathcal{A}\mathcal{Z} + B}{\mathcal{C}\mathcal{Z} + \mathcal{D}} \end{aligned} \tag{6}$$

where the coefficients $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ are constants, and $\mathcal{Z}$ is the complex variable whose phase represents the change in reflection path length. The phase change in the CSI ratio effectively captures the variations in the reflection paths caused by different gestures and the physical characteristics (e.g., length and thickness) of human hands. Thus, the phase change of the CSI ratio encodes both gesture information and user identity. By analyzing these phase changes, it is possible to distinguish between different gestures and identify the individual performing them.

We further implement a visualization method to convert the phase information of the CSI ratio into image representations for classification tasks. This approach capitalizes on the ability of deep learning models to analyze essential information embedded in images. WiFi receivers typically have multiple antennas. To calculate the CSI ratio, we select the two antennas with the most significant differences in received data. Specifically, assuming a single WiFi transmitter, the CSI ratio data from a receiver can be structured into a two-dimensional tensor $M_{[S\times T]}$, where S and T are subcarriers and packet counts. To incorporate data from N receivers, we extend the tensor into a larger two-dimensional matrix $M_{[(N\times S)\times T]}$. This integrated tensor captures both the spatial and temporal dimensions of the signal, providing a comprehensive representation of the CSI data across different

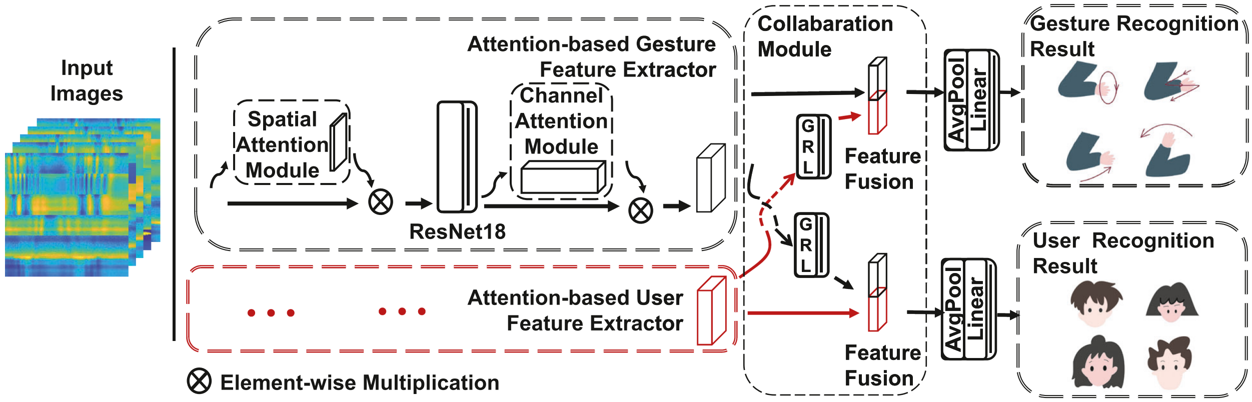


Fig. 3. Overview of the dual-task model architecture. The model integrates attention-based feature extraction modules (spatial attention, ResNet backbone, channel attention) and a collaboration module. The collaboration module incorporates Gradient Reversal Layers to regulate task interactions and feature fusion to enhance gesture and user identity classification before the final output.

receivers. Finally, we use the Matlab function imagesc to convert the two-dimensional matrix into an image [16].

D. Dual-Task Recognition

The multipath effect introduces signal variations even when the same gesture is performed by the same user across different domains, posing challenges for dual-task recognition using CSI ratio features. To address this, we propose a dual-task module (Fig. 3) that leverages an attention-based feature extractor and a collaborative module for adaptive feature extraction in both gesture and user identity recognition tasks. The attention-based feature extraction module integrates spatial and channel attention mechanisms, allowing the model to focus on the most relevant parts of the signal. The collaborative module enhances task performance by fusing features extracted for gesture recognition and user identification.

Attention-based Feature Extractor: In daily tasks, like searching for familiar individuals, our attention naturally focuses on salient elements, such as human faces. The attention mechanism follows a similar principle, directing focus to discriminative image regions processed efficiently through the network module. Leveraging this characteristic, we aim to guide the network to concentrate on pixels essential for recognizing gestures and user identities in signal-converted images. This approach aims to achieve high-precision dual-task recognition across domains.

Basically, the attention mechanism can be formulated as [17]:

$$Attention = f\left(g(x), x\right) \tag{7}$$

The $g(x)$ represents the attention map which corresponds to the process of 'what' and 'where' to pay attention. $f(g(x), x)$ is the process that combines the attention map with the input data x and extracts the important features by emphasizing the regions highlighted by the attention map.

Images inherently encompass information along two fundamental dimensions: spatial and channel axes. Leveraging the attention mechanism in both dimensions assists the network in learning 'what' and 'where' to focus on in an image [17]. The input data consists of images derived from signal-converted data,

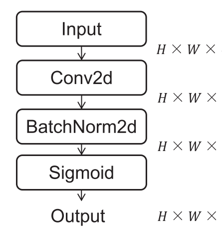


Fig. 4. The spatial attention module.

and we aim to improve dual-task recognition of gestures and user identities. We employ ResNet18 as the backbone of our feature extraction process, augmented with spatial and channel attention modules. ResNet18 [19] is a robust neural network architecture known for its efficacy in classification tasks. We choose ResNet18 for its ability to alleviate gradient vanishing and extract robust features while maintaining computational efficiency. The feature extraction process can be summarized as follows:

$$X' = M_s(X) \otimes X \tag{8}$$

$$X'' = \text{ResNet 18}\left(X'\right) \tag{9}$$

$$X''' = M_c\left(X''\right) \otimes X'' \tag{10}$$

where $X \in R^{C\times H\times W}$ is the input feature map, $\otimes$ denotes element-wise multiplication, M_s denotes the spatial attention module and M_c denotes the channel spatial attention module. We first use the spatial attention module M_s to capture significant spatial information within the image. After spatial attention, the processed data is passed through ResNet18 to further refine and learn complex patterns from the spatially attended data. The output from ResNet18 is then processed through the channel attention module M_c. We further introduce the M_s module and M_c module in detail.

Spatial Attention Module: The Spatial Attention Module focuses on identifying important spatial features relevant to the tasks. We utilize the spatial attention mechanism described in [20]. The structure of this module is detailed in Fig. 4.

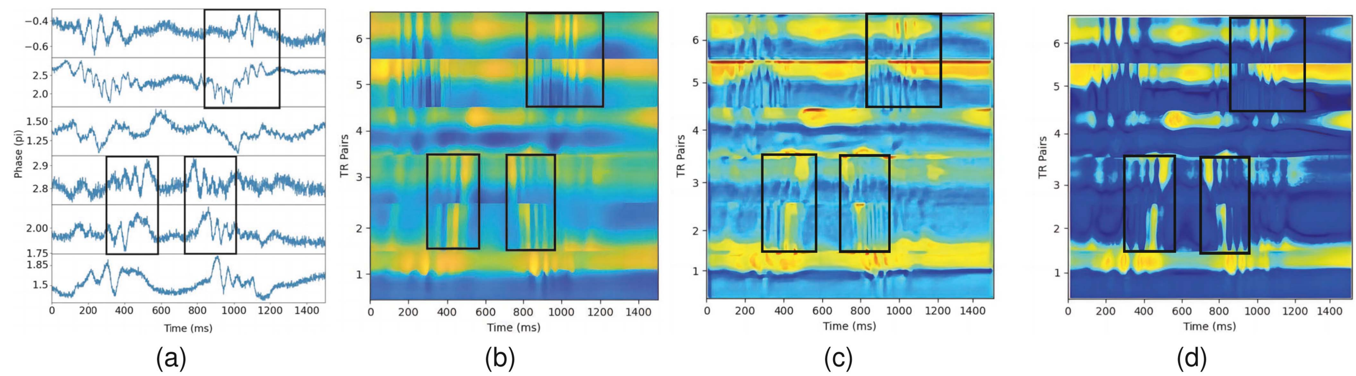


Fig. 5. The function and effectiveness of the attention module. (a) Phase waveforms of different receivers (from top to bottom, the first to sixth receivers). (b) Raw input map. (c) Spatial attention map. (d) Input map after spatial attention map.

The process begins with a convolution layer that uses a kernel size of 7 to create a $2D$ spatial attention map, denoted as $s_s \in \mathbb{R}^{H\times W}$. The convolutional layer in the spatial attention module is indeed specifically designed to preserve the spatial dimensions H and W. This is a deliberate choice to ensure alignment between the input feature map and the spatial attention map, as the latter is applied element-wise to the input feature map. The output from the convolution layer is then passed through a batch normalization layer followed by a sigmoid activation function. The attention map is applied to the tensor of the denoised signal $M_{[(N\times S)\times T]}$. The purpose is to select the important signal regions from the 30 subcarriers across 6 receivers within the time dimension T. Then this module uses the batchnorm layer and sigmoid layer as normalization and activation.

Fig. 5 illustrates the function and effectiveness of the spatial attention module. We use data from a single gesture motion performed by a user as an example for analysis. Fig. 5(a) illustrates the CSI phase waveforms of different antenna pairs after noise reduction, capturing the temporal variations of the channel state. Fluctuations in the CSI phase are closely related to gestures or user-specific characteristics. Certain waveform regions show significant variations, often corresponding to key moments of gesture transitions or unique user traits. We mark several representative segments, rich in information, with black boxes as illustrative examples. Fig. 5(b) visualizes the raw input signal as a time-spatial map, where brighter regions indicate higher signal intensity. By comparing this map with the marked regions in panel (a), the highlighted informative segments in the original signal become clearer. Fig. 5(c) presents the spatial attention map generated from Fig. 5(b) using the spatial attention mechanism. Bright regions in the map correspond to areas assigned higher weights by the attention module, reflecting their greater relevance to the tasks. Fig. 5(d) depicts the final output of the spatial attention module, produced by element-wise multiplication of the raw input (Fig. 5(b)) and the attention map (Fig. 5(c)). This operation amplifies the significant regions while suppressing irrelevant areas, yielding a transformed signal where key features are prominently emphasized. The enhanced brightness within the example boxes visually demonstrates the module's effectiveness in identifying and highlighting critical information relevant to the tasks.

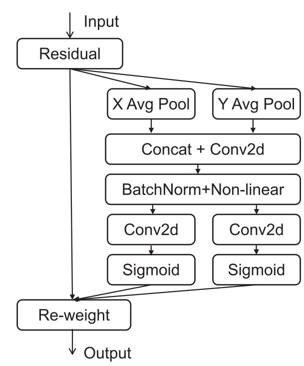


Fig. 6. The channel attention module.

Channel Attention Module: This module is designed to extract an attention map from the channel dimension. After the original data passes through the spatial attention module and ResNet18, its dimensions become $X'' \in R^{C''\times H''\times W''}$. In WiDual, C'' is set to 512, indicating there are 512 channels or feature maps, each of size $H'' \times W''$. These feature maps contain information about specific areas of the original image. The primary goal of this module is to determine "what" areas to focus on, adjusting the weight of each feature map to help the network efficiently concentrate on the most important regions.

Incorporating positional information into the channel attention mechanism allows the network to focus on significant regions with minimal computational expense [17]. To achieve this, we adopt the coordinate attention mechanism for our channel attention module [20]. The structure of this module is shown in Fig. 6. The process begins with horizontal and vertical encoding of each channel through two spatial extents of pooling kernels. The combined outputs from these pooling layers then undergo a shared 1×1 convolutional transformation. This is followed by normalization using a BatchNorm layer and activation using a ReLU layer. The coordinate attention mechanism then splits the resulting tensor into two separate tensors. Each tensor undergoes further convolution and activation to generate attention vectors with the same number of channels. Finally, these attention vectors are multiplied by the input feature maps, thereby enhancing the focus on relevant features.

Collaboration Module: The main goal of this module is to merge features from two tasks–gesture recognition and user identity recognition–to enhance the accuracy of both tasks. The idea is inspired by the observation that different users exhibit diverse habits even when performing the same actions [10]. Therefore, a network's knowledge of user identity can potentially improve its ability to recognize gesture types, and vice versa.

After the initial feature extraction, the collaboration module fuses gesture features extracted by the attention mechanism with user identity features. This fusion takes place before the fully connected layer, which is responsible for the final classification of gestures and user identities. Specifically, the collaboration module applies Gradient Reversal Layers (GRLs) [13] to regulate how features influence the auxiliary task during training. Then, combining the features of both tasks through feature concatenation, ensuring effective feature fusion without mutual interference. The GRL is a crucial part of the collaboration module, facilitating task disentanglement by reversing the gradient flow for one task while preserving it for the other. For example, in gesture recognition, this means that after fusing the features, user identity features are only used as auxiliary features to enhance gesture recognition. The loss from gesture recognition should not affect the optimization of user identity features. As shown in Fig. 3, GRLs are applied to the auxiliary task features before concatenation. By setting the GRL's hyperparameter to zero, we block the gradient flow from one task to the feature extractor of the other, ensuring task-specific feature independence during backpropagation. This mechanism allows the module to achieve effective task collaboration without compromising the individual optimization of each task.

Feature fusion is implemented using PyTorch's concatenate function. For both tasks, we first extract the task-specific features after passing through the attention-based feature extractor. Taking the gesture recognition task as an example, we obtain the gesture recognition features, denoted as out_ges with shape $(batch\_size, C_{ges}, H'', W'')$. Let $reverse\_feature\_user$ denote the feature map generated by the auxiliary task through the GRL. This feature is concatenated with the gesture recognition features to provide richer semantic information for the classifier. After concatenation, the channel dimension is stacked, resulting in a feature map with shape $(batch\_size, C_{ges} + C_{user}, H'', W^{\prime\prime})$, while the spatial dimensions, H'' and W'', remain unchanged. By concatenating along the channel dimension, features from different sources are combined, enhancing the network's ability to focus on important regions from both the gesture and user modules, while preserving spatial location information.

IV. Implementation

A. Model Training Process

The dual-task attention network is designed to handle the complexities of gesture recognition and user identity recognition simultaneously. This network leverages the strengths of residual networks and attention mechanisms to extract features from cross-domain signals. Additionally, through feature fusion, the network leverages the shared information between gesture recognition and user identity recognition tasks, enhancing overall performance.

Algorithm 1: Dual-Task Attention Network Model Training.

Input: Dataset $D^{tr} = \{(x^i, y_1^i, y_2^i)\}_{i=1}^{n_{tr}}$, D^{tr} contains dataset X for the gesture recognition task and dataset X_1 for the identity recognition task
Output: The optimal parameters θ^* of the dual-task recognition model
1: **for** X, X_1 in D^{tr} **do**
2: **while** the model did not converge **do**
3: $X' \leftarrow M_s(X) \otimes X$
4: $X'' \leftarrow \text{ResNet 18}(X')$
5: $X''' \leftarrow M_c(X'') \otimes X''$ // X''' contains multi-scale features $F_g = \{F_{g1}, F_{g2}, \ldots, F_{gn}\}$
6: X_1 repeat steps 3 to 5 to obtain the feature $F_u = \{F_{u1}, F_{u2}, \ldots, F_{un}\}$
7: $F_g' \leftarrow GRL(F_g)$
8: $F_g'' \leftarrow M_{co}(F_g', F_u)$ // M_{co} denotes the collaboration module
9: $X_A \leftarrow \text{Avgpool}(F_g'')$
10: $X_g \leftarrow \text{Linear}(X_A)$
11: F_u repeat steps 7 to 10 to obtain the output X_u
12: $\theta^* \leftarrow L(\theta, D)$ according to (1)
13: Update the model parameters θ^*
14: **end while**
15: **end for**

The training process for the dual-task attention network model is shown in Algorithm 1. Two supervised learning tasks are defined: gesture recognition and identity recognition. Each task has associated training data from the training dataset D^{tr}, data pairs $(x^i, y_1^i)_{i=1}^{n_{tr}}$ for gestures and $(x^i, y_2^i)_{i=1}^{n_{tr}}$ for identities. Lines 3 to 5 extract task-related features from CSI image using the dual-task attention network module. Lines 7 to 8 perform GRL and feature fusion. In line 9, the fused features are passed through an average pooling (Avgpool) layer to reduce its dimensionality while retaining essential information. The pooled features are then processed in line 10 by a linear layer to produce the final task-specific outputs. Line 12 calculates the loss functions for both tasks. Finally, the model parameters are updated through backpropagation (Line 13).

B. Model Parameters

To present the model implementation more clearly, the overall parameter settings of the model are shown in Table II. The spatial attention module starts with a convolutional layer that uses a kernel size of 7. This convolutional layer creates a 2D spatial attention map, which identifies the crucial spatial regions in the input data. Following this, batch normalization and the sigmoid activation function are applied to normalize and activate the map, respectively. ResNet18 is a deep convolutional neural network comprising 18 convolutional layers organized into several residual blocks. In our model, we retain only five convolutional structures from ResNet18, excluding the flatten, softmax, and fully connected layers present in the original model. This modification allows our network to focus on extracting deep-level features from the signals while maintaining a streamlined architecture for efficient processing. The channel attention module employs a 2D convolutional layer with a kernel size of 2. The collaboration module is designed to fuse the features from both tasks, thereby further enhancing the accuracy of each task. The Gradient Reversal Layer (GRL) can reverse the gradients during the training process of the neural network, allowing it to learn both forward and reverse tasks simultaneously. By setting the hyperparameter of GRL to zero, we ensure that the loss from one task does not affect the training of the other task. The final output network layer consists of the flatten layer, fully connected layer, and Sigmoid.

C. Dual-Task Recognition Complexity

The algorithmic complexity of the dual-task recognition network can be primarily analyzed based on the computational cost of its convolutional layers, as these dominate the total time complexity compared to operations like pooling, activation functions, and batch normalization. For a convolutional layer, the time complexity is expressed as $\mathcal{O}(M \times K \times C_{in} \times C_{out})$, where M represents the size of the output feature map, K is the kernel size, C_{in} denotes the number of input channels, and C_{out} is the number of output channels. The size of the feature map M is determined by $M = \frac{size_{Matrix} - size_{Kernel} + 2 \times size_{Pad}}{S}$, with $size_{Matrix}$, $size_{Kernel}$, $size_{Pad}$, and S representing the input matrix size, kernel size, padding size, and stride, respectively. In our dual-task attention-based network, the primary computational cost arises from the spatial and channel attention modules as well as the backbone network (ResNet18). Each spatial attention module includes a convolutional layer with parameters $M = 224, K = 1, C_{in} = 512, C_{out} = 16$, resulting in a complexity of 1.83×10^6. The channel attention module consists of three convolutional layers with complexities of 1.83×10^6, 2.93×10^7, and 2.93×10^7, totaling 6.04×10^7. The backbone ResNet18 contributes 1.8×10^9 to the complexity. Combining these components, each feature extractor has a total complexity of 1.86×10^9, and the complete dual-task model (with two feature extractors) has a total time complexity of 3.7×10^9. This analysis highlights that the model achieves efficient computation despite its dual-task nature, ensuring feasibility for real-time applications.

V. Evaluation

Similar to state-of-the-art dual-task systems [9], [10] we evaluate the performance of WiDual using the Widar3 dataset [4]. First, we provide the dataset configuration and implementation details. We then assess the effectiveness of WiDual by comparing its performance against state-of-the-art methods across three dimensions: in-domain performance, cross-domain performance, and latency. Lastly, we examine the contributions of the spatial attention module, the channel attention module, and the collaboration module within the attention-based feature extractor.

A. Setup

Dataset Configuration: We conduct experiments on the Widar3 dataset [4] which is a WiFi dataset comprising 20 different gestures performed by 15 users. However, the dataset is imbalanced: some users perform only 6 gestures, while others perform up to 15. For our verification experiments, we selectively use the data based on specific needs. The detailed configuration for dual-task recognition is provided in Table I. For each gesture performed by each user, the number of signal samples is calculated as $L \times O \times Rep \times Rec$, where L represents the number of Locations, O denotes the number of orientations, Rep signifies the number of repetitions of the gesture, and Rec indicates the number of WiFi receivers. For instance, one user in the first row of Table I performs 9 gestures in 5 locations and 5 orientations, each repeated 5 times, resulting in $5 \times 5 \times 9 \times 5 = 1125$ samples per WiFi receiver. With 6 receivers and 9 users, the total sample size for this row is $1125 \times 6 \times 9 = 60{,}750$ samples.

Implementation Details: We train and implement our model on a PyTorch 1.8 platform, running on a server with Ubuntu 20.04 LTS and powered by an RTX 3090Ti-24G GPU. The training process utilizes a learning rate of 0.001 and a batch size of 20. We allocate 80% of the data for training and 20% for testing, and conduct five-fold cross-validation for in-domain, cross-location, and cross-orientation evaluations. For in-domain evaluation, we test one repetition and train on the remaining four. Cross-orientation evaluation involves testing one orientation while training on the other four. Similarly, cross-location evaluation tests one location with the other four used for training. For cross-environment evaluation, due to dataset limitations, we use data from environments 1 and 2 for training and data from environment 3 for testing.

Metrics: To evaluate the performance of WiDual and compare it with Widar3, WiHF [10], and WiGestID [9], we use several key metrics: latency, accuracy (ACC), false accept rate (FAR), and false reject rate (FRR). These metrics are applied to both gesture recognition and user identification tasks. ACC measures the proportion of correct predictions out of the total predictions made by the system. It provides an overall indication of the system's confidence and reliability in its predictions. FAR measures the likelihood that the system incorrectly accepts an unauthorized user. A lower FAR is desirable as it indicates fewer instances of security breaches. FRR indicates the probability that the system incorrectly rejects an authorized user. A lower FRR is preferable as it indicates fewer instances of denying legitimate access. Using the confusion matrix, we can calculate the metrics as follows:

$$ACC = \frac{TruePositive}{TruePositive + FalsePositive} \tag{11}$$

$$FAR = \frac{FalsePosition}{FalsePosition + TrueNegative} \tag{12}$$

$$FRR = \frac{FalseNegative}{FalseNegative + TruePositive} \tag{13}$$

TABLE I
DESCRIPTION OF THE DATASET WE USE

Environments	User	Gesture	No. of Loc[1]	No. of Ori[1]	No. of Rep[1]	No. of Sam[1]
Classroom	user1, user2, user3, user10-15	1:Push&Pull; 2:Sweep; 3:Clap; 4:Slide; 5:Draw-O; 6:Draw-Zigzag;7:Draw-N; 8:Draw-Triangle;9:Draw-Rectangle;	5	5	5	60750
Hall	user1, user2, user3	1:Push&Pull; 2:Sweep; 3:Clap; 4:Slide; 5:Draw-O (Horizontal); 6:Draw-Zigzag (Horizontal);	5	5	5	13500
OfficE	user3	1:Push&Pull; 2:Sweep; 3:Clap; 4:Slide; 5:Draw-O (Horizontal); 6:Draw-Zigzag (Horizontal);	5	5	5	4500

[1] No. of Loc: number of locations; No. of Ori: number of orientations; No. of Rep: number of repetitions; No. of Sam: number of samples.

TABLE II
WIDUAL MODEL PARAMETER SETTINGS

Module		Network Layer[1]	Output
Input		/	$10 \times 3 \times 224 \times 224$
Spatial Attention Module	Task1	Conv2d(Convolutional kernel: 7×7, Padding: 3)+BN+Sigmoid	$10 \times 512 \times 7 \times 7$
	Task2	Conv2d(Convolutional kernel: 7×7, Padding: 3)+BN+Sigmoid	$10 \times 512 \times 7 \times 7$
ResNet18 Module	Task1	ResNet18 without the final Flatten and Softmax layers	$10 \times 512 \times 7 \times 7$
	Task2	Consistent with Task1	$10 \times 512 \times 7 \times 7$
Channel Attention Module	Task1	As shown in Fig. 5, Conv2d(Convolutional kernel:1×1,Padding: 0)	$10 \times 512 \times 7 \times 7$
	Task2	Consistent with Task1	$10 \times 512 \times 7 \times 7$
Collaboration Module	Task1	GRL(hyperparameter:0)+Concat	$10 \times 1024 \times 7 \times 7$
	Task2	Consistent with Task1	$10 \times 1024 \times 7 \times 7$
Output	Task1	Flatten+FC+Sigmoid	10×6
	Task2	Consistent with Task1	10×6

[1] Conv2d: 2D Convolutional Layer; BN: Batch Normalization Layer; Sigmoid: Activation Layer; Flatten: Flattening Layer; Softmax: Activation Layer; GRL: Gradient Reversal Layer; FC: Fully Connected Layer.

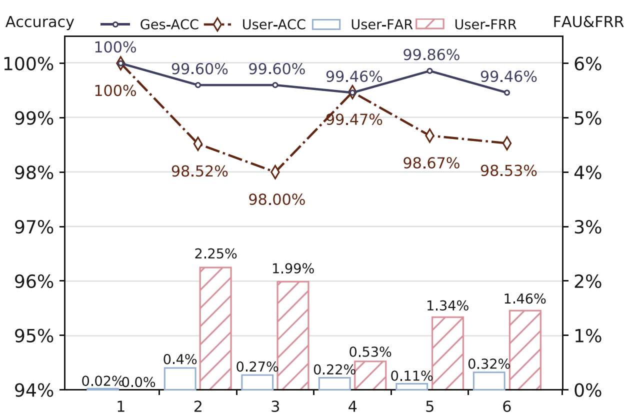


Fig. 7. Overall performance of WiDual. Target Index denotes the specific gesture and user target for dual tasks.

B. Overall Performance

To evaluate system performance across various domains, including all orientations, locations, and environments, we use 80% of the dataset for training and 20% for testing. WiDual achieves a gesture recognition accuracy of 99.67% and a user identification accuracy of 98.87%. Fig. 7 presents the overall performance metrics, showing ACC of gesture and user recognition, as well as FAR and FRR for user identification. The horizontal axis in the figure represents the specific labels for gestures and user identities in the dual-task scenario. Notably, WiDual consistently achieves over 98% accuracy for all 6 gestures and 6 users across various domains.

Furthermore, we compare the overall performance of WiDual with state-of-the-art methods. As shown in Table III, WiDual's gesture recognition accuracy surpasses those of Widar3 (92.7%), WiHF (97.65%) and WiGestID (98.51%). For user identity recognition, WiDual achieves an accuracy of 98.87%. This demonstrates WiDual's capability to accurately identify both gestures and use identities offers advantages in dual-task recognition scenarios.

TABLE III
ACCURACY COMPARISONS WITH STATE-OF-THE-ART METHODS

	Gesture Recognition	User Identification
Widar3 [4]	92.7%	/
WiHF [10]	97.65%	96.74%
WiGestID [9]	98.51%	97.32%
WiDual	99.67%	98.87%

C. Cross-Domain Evaluation

To evaluate the cross-domain accuracy of WiDual, we first measure its performance in gesture recognition and compare it with Widar3 and WiHF. Using five-fold cross-validation, we calculate the average accuracy across various domains. While assessing a specific domain component, other domain components are kept constant. The accuracy distribution for each domain component is shown in Fig. 8. Compared to Widar3 and WiHF, WiDual shows an increase in average gesture recognition accuracy across different locations, orientations, and environments. Overall, in the gesture recognition task, Widar3 achieves a cross-domain performance of 86.39%, WiHF achieves 89.46%, and WiDual achieves 96%. Note that the cross-orientation accuracy of WiDual drops the most compared to its in-domain performance, with an average accuracy decreased from 99.67% to

Fig. 8. Accuracy comparison between WiDual, Widar3, and WiHF in terms of gesture recognition across domains.

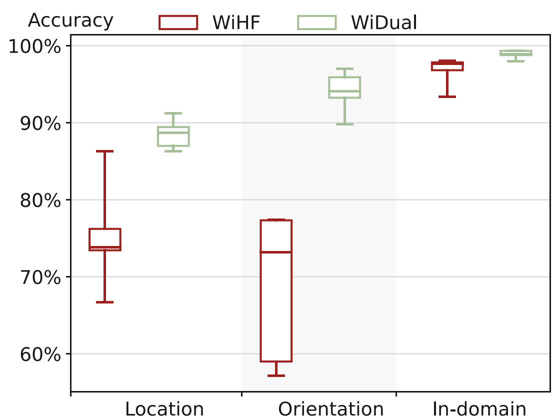


Fig. 9. Accuracy comparison between WiDual and WiHF in terms of user identity recognition across domains.

94.91%. This decline is similar to that was observed with Widar3 and WiHF and can be attributed to signal path obstruction by the human body at certain orientations, leading to information loss in the received signal.

Fig. 9 illustrates the comparison of user identity recognition accuracy between WiDual and WiHF across different domains. The overall average accuracy of WiHF is 72.05%, while WiDual achieves 91.27%, representing a 26% improvement. We further break down the performance of user identity recognition at different locations and different orientations in Fig. 10. We can see that WiDual maintains higher accuracy compared to WiHF across various locations and orientations. The key factor behind this disparity is WiHF's reliance on manual feature extraction, which can omit crucial details such as torso variations affecting signal characteristics. In contrast, WiDual leverages an attention-based feature extraction network that autonomously learns and identifies relevant features for the recognition task. However, both WiDual and WiHF show decreased accuracy in directions 1 and 5. This reduction is due to increased signal obstruction by the human body in these directions, making it difficult for the receivers to capture adequate information on body movements, thus hindering accurate user identity recognition.

TABLE IV
LATENCY COMPARISONS WITH STATE-OF-THE-ART METHODS

	Widar3	WiGestID	WiHF	WiDual
Signal Preprocessing	0.162s	0.162s	1.557s	0.364s
Feature Extraction	70.29s	70.29s	0.379s	/
Total Time Consumption[1]	70.61s	>70.452s	2.488s	0.379s

[1] Loading data, signal processing, feature extraction, and recognition and identification are all involved.

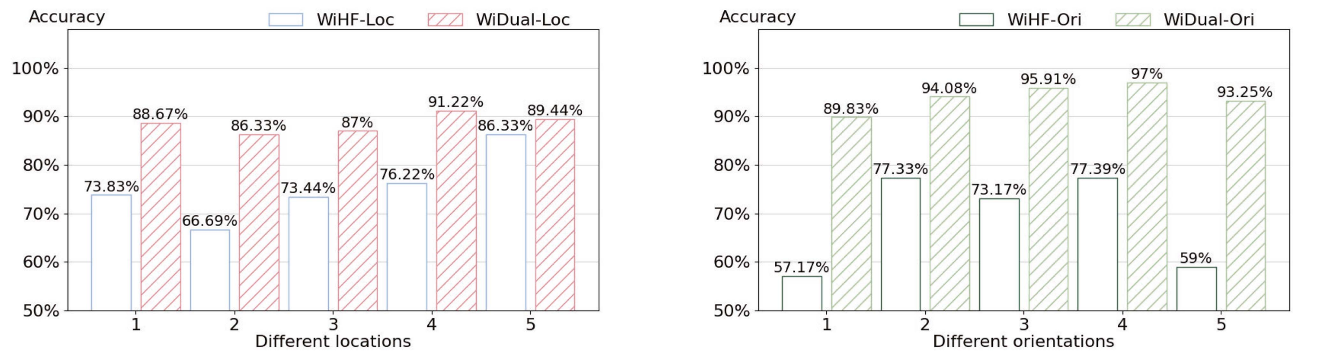
D. System Insights

In this section, we conduct a comprehensive evaluation of WiDual's performance, focusing on three key aspects: latency, the impact of each module, and the effect of increasing the number of gestures and users.

1) Latency: Latency is a critical factor for user experience, so we measure the processing time of WiDual and compare it with other systems. The latency here refers to the time required to execute a single dual-task recognition process. This includes both gesture recognition and user identity recognition. Specifically, the latency captures the time taken for the system to process one gesture or action from data acquisition to final classification results for both tasks. The total response time of WiDual primarily involves CSI preprocessing and passing data through the network module. Table IV illustrates the time distribution across these stages. Unlike other systems, WiDual significantly reduces the time spent before data enters the network. For instance, WiGestID relies on the BVP feature used in Widar3, which involves searching in a high-dimensional space, making it time-consuming and dependent on the fixed locations of WiFi receivers. If the receiver locations change, the BVP features must be recalculated, extending feature extraction time. In contrast, WiHF uses the seam carving algorithm for more efficient feature extraction, reducing processing time by 30 times compared to Widar3. WiDual further improves efficiency, reducing processing time by 6 times compared to WiHF. Since the processed tensor $M_{[(N\times S)\times T]}$ can be derived from different receivers and subcarriers independently and concatenated together, the results are processed in parallel across receivers.

2) Module Evaluation: We evaluate the effectiveness of WiDual when integrated with different modules: the spatial attention module, the channel attention module, and the collaboration module. As shown in Table V, while the modules did not significantly enhance in-domain performance due to the powerful pre-trained ResNet18 model, they proved beneficial in cross-domain scenarios. The spatial and channel attention modules each improved accuracy by approximately 2% for dual tasks. Building on the attention modules, the collaboration module can further enhance performance.

3) Impact of Different Amounts of Training Data: We conduct experiments to understand how varying amounts of training data affect the performance of our dual-task model in in-domain, cross-location, and cross-direction scenarios. We train the model using 1/4, 1/2, and 3/4 of the available training data, respectively, and then evaluate its performance using a consistent test set. As shown in Table VII, results indicate that the model's accuracy



(a) User identity recognition at different locations. (b) User identity recognition at different orientations.

Fig. 10. Accuracy comparison between WiDual and WiHF for each user in terms of user identity recognition across domains. WiHF-Loc and WiHF-Ori represent WiHF's recognition performance across locations and orientations, respectively, while WiDual-Loc and WiDual-Ori denote WiDual's performance in the same aspects. (a) Cross-Location. (b) Cross-Orientation.

TABLE V
ACCURACY WITH DIFFERENT MODULES

	IN_G[1]	IN_U[1]	CO_G[1]	CO_U[1]	CL_G[1]	CL_U[1]
WiDual	99.67%	98.87%	94.91%	94.31%	97.24%	88.53%
WiDual-S[2]	99.42%	98.33%	93.93%	93.13%	95.78%	85.6%
WiDual-C[2]	99.62%	98.91%	93.33%	94.18%	96.13%	87.06%
WiDual-Co[2]	99.68%	98.69%	94.33%	93.6%	96.44%	87.28%

[1] IN_G: gesture recognition in-domain; IN_U: user identification in-domain; CO_G: gesture recognition cross-orientation; CO_U: user identification cross-orientation; CL_G: gesture recognition cross-location; CO_L: user identification cross-location.

[2] WiDual-S: WiDual without Spatical Attention Module; WiDual-C: WiDual without Channel Attention Module; WiDual-Co: WiDual without Collatoration Module.

TABLE VI
IMPACT OF GESTURE AND USER TYPE NUMBERS

The number of Gestures		6	7	8	9
Gesture Recognition	In-domain	99.67%	99.54%	99.08%	99.15%
	Orientation	94.91%	95.04%	91.01%	91.17%
	Location	97.24%	97.31%	94.7%	95.11%
User Identification	In-domain	98.87%	98.91%	98.95%	98.97%
	Orientation	94.31%	94.15%	94.1%	94.7%
	Location	88.53%	88.34%	88.68%	89.33%
The number of Users		**6**	**7**	**8**	**9**
Gesture Recognition	In-domain	99.67%	99.75%	99.63%	99.55%
	Orientation	94.91%	95.00%	95.71%	94.87%
	Location	97.24%	97.33%	97.7%	97.39%
User Identification	In-domain	98.87%	98.97%	98.21%	98.44%
	Orientation	94.31%	93.16%	90.7%	91.34%
	Location	88.53%	85.33%	80.5%	81.33%

is influenced by the amount of training data. Specifically, when the training data is reduced to 1/4 of the original amount, the average accuracy for both gesture recognition and user identity recognition in cross-domain scenarios drops to around 80%. This reduction in performance is due to the limited data available, which restricts the network's ability to learn the necessary distinguishing features for high accuracy.

TABLE VII
IMPACT OF DIFFERENT AMOUNTS OF TRAINING DATA

Training Sets		1/4	1/2	3/4	1
Gesture Recognition	In-domain	93.93%	98.97%	99.53%	99.67%
	Orientation	79.13%	88.73%	94.18%	94.91%
	Location	82.75%	92.60%	95.56%	97.24%
User Identification	In-domain	93.71%	97.15%	98.28%	98.87%
	Orientation	85.69%	90.96%	92.04%	94.31%
	Location	75.36%	82.18%	85.27%	88.53%

1/4 means use 1/4 of all training data.

4) Impact of Number of Gestures and Users: To analyze the effect of the number of gestures and user types on the performance of WiDual, we divide the dataset into several subsets, maintaining a default of 6 types for both tasks. As shown in Table VI, the in-domain accuracy for gesture recognition stays above 99%, and user identification accuracy remains above 98%, even when the number of gestures or users increases to 9. This indicates that WiDual's accuracy is not significantly affected by the increase in the number of gestures or users. In some instances, increasing the number of gestures or users actually enhances accuracy, possibly because the newly added gestures or users are easier to distinguish. Additionally, an increase in the number of gestures does not compromise the accuracy of user identification, and vice versa. This stability in performance is primarily attributed to the gradient reversal layer within the collaboration module.

5) Impact of the Number of WiFi Receivers: We examine how varying the number of WiFi receivers affects the performance of WiDual. The results are shown in Table VIII. With only two receivers, the model maintains high accuracy for both gesture recognition and user identification in the in-domain scenario. However, in cross-direction and cross-location scenarios, performance becomes more sensitive to the number of receivers. This sensitivity occurs because additional receivers provide more comprehensive directional information, enabling the network to learn richer features. A reduction in the number of receivers has a more significant impact on user identification accuracy than on gesture recognition. Specifically, when the number of receivers

TABLE VIII
IMPACT OF THE NUMBER OF WIFI RECEIVERS

WiFi receivers		2	3	4	5	6
Gesture Recognition	In-domain	99.53%	99.60%	99.49%	99.51%	99.67%
	Orientation	92.80%	92.80%	92.87%	94.07%	94.91%
	Location	96.71%	96.95%	96.93%	96.89%	97.24%
User Identification	In-domain	97.69%	97.53%	97.78%	98.27%	98.87%
	Orientation	89.02%	89.96%	90.02%	90.89%	94.31%
	Location	81.62%	82.38%	82.93%	85.11%	88.53%

is reduced from six to two, gesture recognition accuracy across different directions decreases by 2.11%, while user identification accuracy drops by 5.29%.

VI. Future Work and Discussion

WiDual focuses on real-time and high-precision user gesture recognition and identity authentication in cross-domain scenarios using WiFi channel state information. There are still limitations that warrant further investigation and improvement.

Effectively recognizing undefined gestures: WiDual currently is capable of recognizing nine predefined user gestures. It remains challenging to enable users to define their own gestures to enhance user experience and practicality. This presents a challenge that may require collecting a limited number of samples for each user-defined gesture. Future research may focus on developing methods to robustly recognize these custom gestures with minimal sample sizes.

Expanding the dataset to include diverse personnel types: The experiments and validations in this study utilize the largest publicly available WiFi perception dataset, but its limited number of volunteers poses certain constraints. To improve the dataset's applicability to real-life scenarios, future work should involve expanding the dataset to include more application scenarios and systematically collecting data that represents a broader spectrum of gesture habits, age groups, and demographics.

Extending to multi-user dual-task recognition: Extending the system to support multi-user simultaneous dual-task recognition is a challenging problem due to several factors: interference in CSI signals, feature decoupling complexity, and computational overhead. When multiple users are present, their gestures introduce overlapping and interfering patterns in the CSI data, significantly increasing the complexity of signal analysis and feature extraction. Besides, multi-user scenarios require the system to disentangle gesture-specific and user-specific features for each individual simultaneously. This increases the difficulty of preserving task-specific features without mutual interference. We plan to explore these challenges in future work.

Exploring alternative loss functions for improved performance: Although cross-entropy loss meets our current requirements, we recognize that alternative loss functions, such as focal loss, may provide benefits in specific scenarios, such as addressing class imbalance or improving performance on underrepresented classes. Focal loss, for instance, places greater emphasis on harder-to-classify samples, potentially enhancing model performance in imbalanced datasets.

VII. Related Works

User Identity Recognition Using WiFi: Several studies have explored using Channel State Information (CSI) to identify human identity based on gait [21], [22], [23], [24], [32], behaviors [25], or body state [8], [9]. WiFiU [21], WiFi-ID [22], and WiAU [23] employ deep learning and other techniques to extract distinctive features from CSI variations associated with walking patterns. GaitFi [32] combines WiFi and camera data to enhance gait-based user identity recognition. LW-WiID [31] improves recognition accuracy by extracting spatial information from subcarriers using a lightweight deep learning algorithm. Unlike these prior works, WiDual is designed to achieve dual-task recognition by simultaneously recognizing user identity and user gestures.

Gesture Recognition Using WiFi: Systems designed for WiFi-based gesture recognition require adaptive capabilities to accommodate new data domains effectively. Widar3 [4] extracts the domain-independent feature BVP from CSI but is sensitive to the accuracy of transceiver location, making it vulnerable to noise and outliers [10]. WiFine [33] offers a real-time gesture recognition system tailored for edge devices, providing quick and accurate results without requiring high-performance services. GWrite [34] can recognize gesture shapes composed of straight-line segments through walls. However, these methods face challenges in supporting real-time processing and preserving user identity information. Different from these works, WiDual performs simultaneous gesture and user recognition to enhance system security and user experience.

Dual-task Recognition Using WiFi: WiID [12] is the pioneering WiFi-based dual-task system but does not support cross-location and cross-environment recognition. WiHF [10] introduces an alternative domain-independent feature called motion change pattern for dual tasks. However, this manually crafted feature struggles to capture sufficient cross-domain information, posing challenges in maintaining high accuracy across various domains. WiGesID [9] achieves dual tasks by identifying new users and recognizing gestures with small sample sizes but relies on Widar3's time-consuming BVP feature, limiting its real-time capabilities. Different from these works, WiDual utilizes CSI vitalization method and employs dual-task attention network and collaborative module to achieve real-time processing and robust cross-domain performance.

VIII. Conclusion

In this paper, we present WiDual, an advanced real-time dual-task system designed for precise cross-domain gesture recognition and user identification using WiFi signals. WiDual leverages an attention mechanism to dynamically identify and prioritize salient features for both tasks, effectively adapting to different domains. To facilitate real-time processing, we employ a CSI visualization technique that transforms WiFi signals into images, which allows for efficient feature extraction and model training. This approach helps avoid the potential information loss and delays associated with the direct extraction of handcrafted features from raw WiFi signals. WiDual further enhances dual-task recognition by incorporating a collaboration module that combines gesture and user identity features. This module ensures that the system benefits from the combined information, improving overall performance. We implement WiDual and evaluate it using the Widar3 public dataset, and the results demonstrate its superior performance compared to existing state-of-the-art methods. WiDual achieves a 26% improvement in cross-domain user identification accuracy and an 8% improvement in gesture recognition accuracy. For in-domain dual tasks, WiDual achieves an accuracy of 99.67% for gesture recognition and 98.87% for user identification.

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